

System1 classifier comparisons for customer-care transcripts

Open encoders · TypeSafe.ai JEV · GPT-5.6 Sol and Luna

99.93%

Best accuracy

GPT-5.6 Sol

99.90%

JEV calibrated

4 / 4,050 errors

99.16%

Best local encoder

ModernBERT trained

1,000 synthetic conversations · 27 attributes · 700 train / 150 validation / 150 locked test

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Executive findings

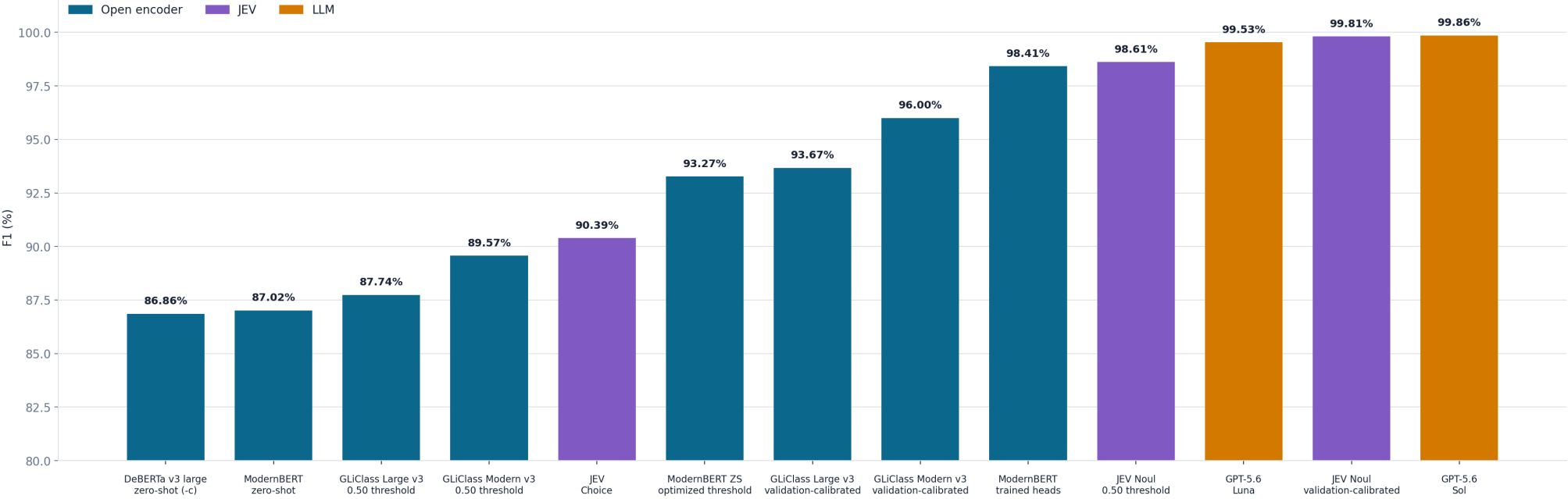
Quality, calibration, and operating-cost implications

- Sol achieved the highest measured quality: 99.93% accuracy and 99.86% micro F1, with three errors across 4,050 held-out decisions.
- Validation-calibrated JEV Noul was one decision behind Sol: 99.90% accuracy and 99.81% F1. Its estimated recurring API charge is about \$0.117 per 1,000 transcripts under the measured prompt shape.
- The strongest fully local result was frozen ModernBERT plus 27 supervised logistic heads: 99.16% accuracy and 98.41% F1.
- Threshold calibration alone improved unchanged zero-shot ModernBERT from 92.96% to 96.42% accuracy and cut errors from 285 to 145.
- GLiClass Modern Large v3 was the strongest zero-shot open encoder: validation-only calibration reached 97.90% accuracy and 96.00% F1.
- DeBERTa-v3-large zero-shot (-c) did not beat ModernBERT zero-shot and was approximately 2.47x slower on the measured CPU path.
- At the 224-token benchmark mean, zero-shot ModernBERT is cheaper than JEV; above about 586 state tokens, JEV Noul becomes cheaper per transcript under the stated H100 assumptions.

Interpret carefully. This is a synthetic, template-driven smoke test. It is useful for controlled comparisons and pipeline validation, but it does not establish production accuracy. The JEV Choice and Noul runs also differ in state structure and criteria specificity, so their gap is not a primitive-only A/B result.

F1 comparison across transcript classifiers

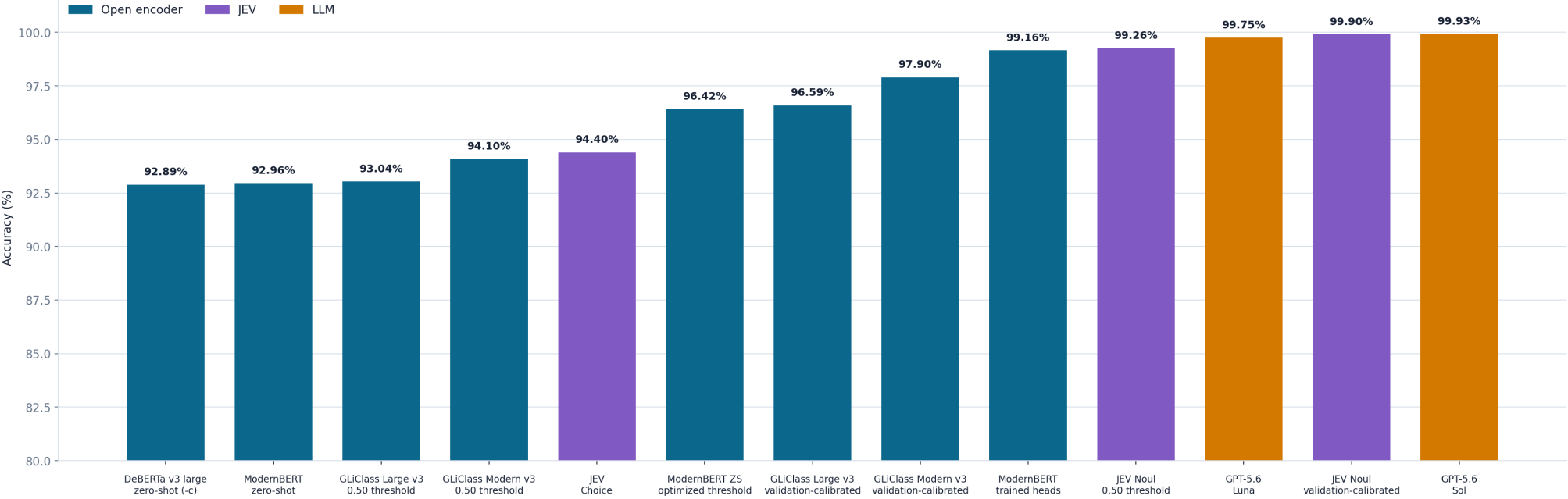
150 locked test transcripts · 27 binary attributes · models ordered by accuracy



F1 axis begins at 80% to make differences legible.

Accuracy comparison across transcript classifiers

150 locked test transcripts · 27 binary attributes · models ordered by accuracy



Accuracy axis begins at 80% to make differences legible.

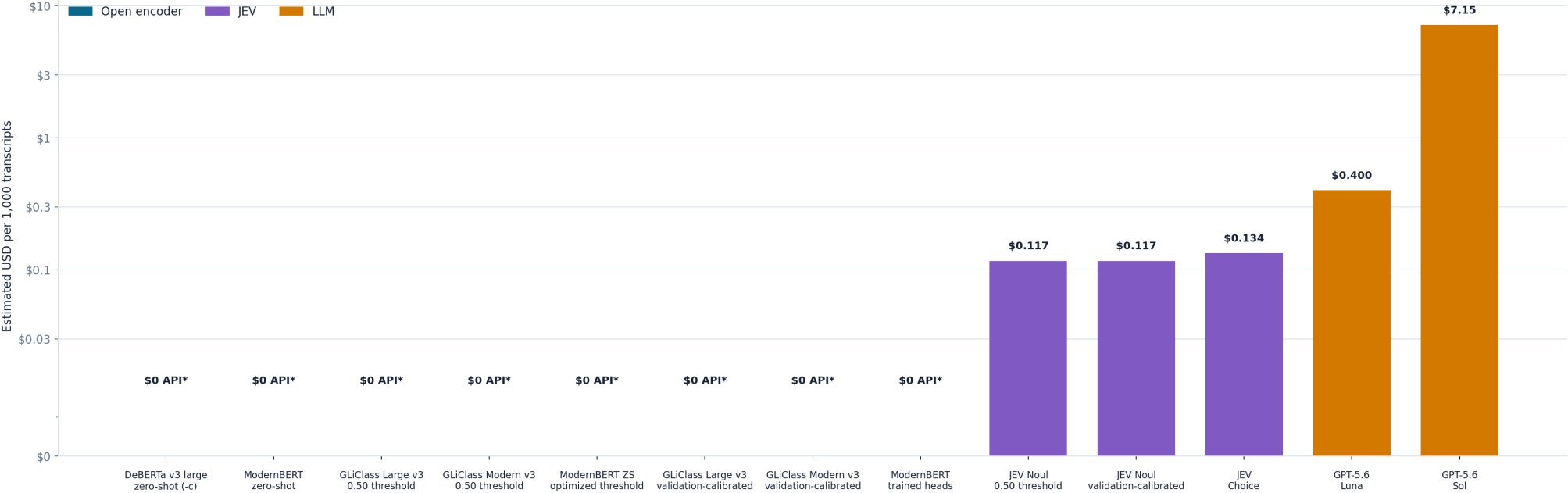
All transcript-classification quality metrics

Held-out test set · micro metrics across 4,050 decisions; exact match requires all 27 labels correct

Approach	Family	Accuracy	Precision	Recall	F1	Exact match
DeBERTa v3 large zero-shot (-c)	Open encoder	92.89%	84.70%	89.14%	86.86%	13.33%
ModernBERT zero-shot	Open encoder	92.96%	84.74%	89.42%	87.02%	14.00%
GLiClass Large v3 0.50 threshold	Open encoder	93.04%	81.90%	94.48%	87.74%	12.00%
GLiClass Modern v3 0.50 threshold	Open encoder	94.10%	83.89%	96.07%	89.57%	16.67%
JEV Choice	JEV	94.40%	82.47%	100.00%	90.39%	18.00%
ModernBERT ZS optimized threshold	Open encoder	96.42%	92.53%	94.01%	93.27%	34.67%
GLiClass Large v3 validation-calibrated	Open encoder	96.59%	91.82%	95.60%	93.67%	34.67%
GLiClass Modern v3 validation-calibrated	Open encoder	97.90%	96.41%	95.60%	96.00%	53.33%
ModernBERT trained heads	Open encoder	99.16%	98.05%	98.78%	98.41%	83.33%
JEV Noul 0.50 threshold	JEV	99.26%	97.27%	100.00%	98.61%	80.67%
GPT-5.6 Luna	LLM	99.75%	99.07%	100.00%	99.53%	93.33%
JEV Noul validation-calibrated	JEV	99.90%	99.63%	100.00%	99.81%	97.33%
GPT-5.6 Sol	LLM	99.93%	99.72%	100.00%	99.86%	98.00%

Estimated marginal inference cost for 1,000 transcripts

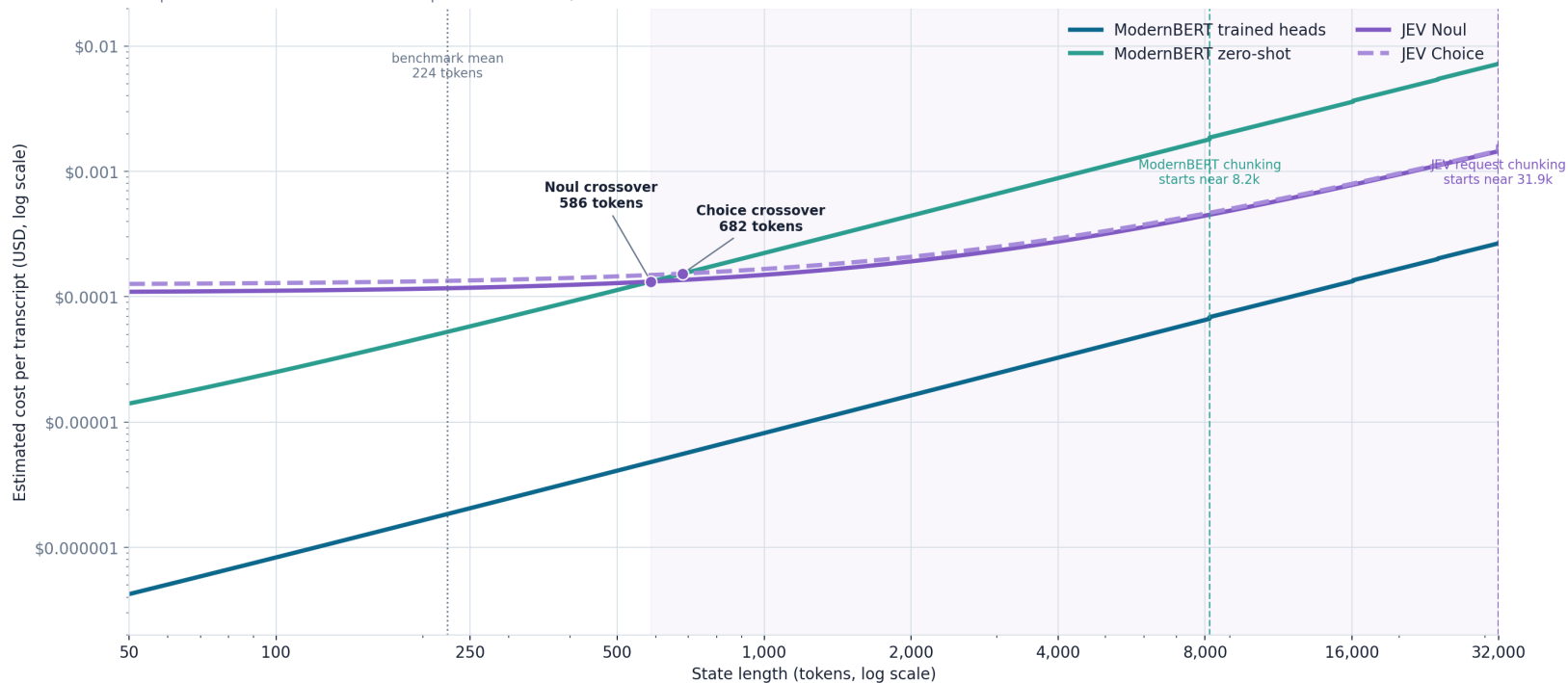
USD API charges · logarithmic scale above \$0.03 · validation/training and local hardware excluded



* Local encoders have no API fee; electricity, hardware, hosting, and operations are not zero and are not estimated. Sol/Luna use a standardized compact prompt estimate and exclude hidden reasoning tokens.

Estimated cost per transcript by state length

27 questions · 256-token chunk overlap · H100 at USD 5/hour · lower cost is better



Each curve is the estimated cost of processing one transcript. Chunking repeats the 256-token overlap and question/request overhead; aggregation compute is excluded. ModernBERT throughput is held at 170.3k processed tokens/s; tokenizer and long-context throughput differences are not modeled.

Cost formulas and assumptions

Dollar cost per transcript - not normalized cost per token

Definitions

S = state tokens q = 27 questions overlap = 256 tokens

$kM = \max(1, \text{ceil}((S - 256) / (8178.11 - 256)))$ $kJ = \max(1, \text{ceil}((S - 256) / (31890 - 256)))$

ModernBERT trained heads $0.008154 * [S + 256*(kM - 1) + 2*kM] / 1,000,000$

ModernBERT zero-shot $0.008154 * \{27*[S + 256*(kM - 1)] + 375*kM\} / 1,000,000$

JEV Noul $0.042 * [S + 256*(kJ - 1) + 2559.74*kJ] / 1,000,000$

JEV Choice $0.042 * [S + 256*(kJ - 1) + 2963.67*kJ] / 1,000,000$

Rates. The H100 scenario converts USD 5/GPU-hour and 170.3k processed tokens/second into USD 0.008154 per million ModernBERT processed tokens. JEV uses USD 0.042 per million billed input tokens.

GLiClass. The new checkpoints appear in the API-cost chart as USD 0 API fee. An H100 cost curve is not estimated because these runs measured CPU inference; the general formula is measured H100 seconds per transcript * USD 5 / 3,600.

What is shared - and what is fixed

The low-cost trained encoder and arbitrary-question JEV solve different serving problems

ModernBERT paths

Zero-shot NLI: N state/question pairs. Measured token amplification: 28.67x for 27 questions.

- It accepts arbitrary hypotheses, but repeats the state inside each premise/hypothesis sequence.
- Trained heads encode the state once, then apply 27 logistic functions.
- The trained path has no question tokens at inference; label meaning is stored in fixed head weights.
- Above about 8.2k state tokens, both paths require overlapping chunks and probability aggregation.
- A novel question requires a new or retrained head.

JEV evidence

- The API accepts arbitrary state and arbitrary typed questions together.
- A state can approach 32k tokens minus its longest question; larger states require multiple requests.
- Billing behaves like state + N x question + fixed overhead.
- Latency was approximately flat from 1 to 16 questions.
- TypeSafe claims a new architecture, parallel sampler, and RLCD training.
- Public materials do not disclose the compute graph or prove state is encoded once.

Conclusion: JEV's arbitrary-question, shared-state-like product behavior is real and useful. It is not enough to verify architectural novelty. Ordinary GPU batching can hide repeated state compute at small N, and known dual-encoder, cached-state, late-interaction, or shared-memory designs could expose similar behavior.

Methodology and controls

What was held fixed and what changed

Dataset

- 1,000 synthetic alternating Agent/Caller conversations.
- 27 binary attributes per transcript.
- Scenario-family-stratified split: 700 / 150 / 150.
- Test truth excluded from training, prompt refinement, and threshold selection.

Metrics

- Micro accuracy, precision, recall, and F1 over 4,050 decisions.
- Exact match requires all 27 labels in a transcript to be correct.

Encoder variants

- Zero-shot: fixed premise/hypothesis NLI probabilities.
- GLiClass: uni-encoder label scoring; Modern v3 used one pass, while the 512-token Large v3 used two label groups to prevent truncation.
- Optimized threshold: 27 validation-selected scalar cutoffs; no weight training.
- Trained: frozen 1,024-d embedding plus 27 logistic heads and validation thresholds.

JEV and LLM variants

- JEV sends 27 typed questions in one request per transcript.
- Final Noul uses structured speakers and strict true/false boundaries.
- Sol and Luna produced blind binary JSON predictions before truth was joined.

Key distinction: trained ModernBERT learned approximately 27,675 logistic-head parameters from 700 rows. The optimized-threshold variant learned only 27 cutoffs from validation. The difference between their test results therefore estimates the value of supervised decision boundaries beyond operating-point calibration.

Conclusions, limitations, and sources

Practical interpretation

- For the highest measured quality, Sol led by one decision over calibrated JEV Noul.
- JEV Noul combined near-frontier quality with arbitrary runtime questions; its estimated per-transcript cost crosses below zero-shot ModernBERT at about 586 state tokens under the stated H100 assumptions.
- Trained ModernBERT was by far the cheapest path, but only because its 27 questions were converted into fixed supervised heads.
- GLiClass Modern Large improved the calibrated zero-shot open-encoder result to 97.90% accuracy, but remained below trained ModernBERT.
- Threshold calibration is worthwhile, but it does not replace supervised heads when enough labeled examples exist.

Limitations

- Synthetic templates are more explicit and regular than production conversations.
- Only 150 transcripts were held out; attribute decisions within a transcript are correlated.
- Choice versus Noul changed multiple factors and is not a clean primitive-only comparison.
- The H100 scenario uses a throughput proxy and holds token throughput constant across state lengths, not a benchmark of this exact checkpoint and serving stack.
- JEV cost is extrapolated from billing; hosted capacity at the normalized throughput was not tested.
- Sol/Luna standardized cost estimates exclude unavailable hidden reasoning-token usage.
- GLiClass Large used two label groups while GLiClass Modern Large used one; their local timings are not a one-pass checkpoint comparison.

Sources

- TypeSafe API: <https://docs.typesafe.ai/api>
- TypeSafe model limits: <https://docs.typesafe.ai/models>
- TypeSafe JEV launch/pricing: <https://typesafe.ai/blog/introducing-system-one-models-and-jev>
- ModernBERT documentation: https://huggingface.co/docs/transformers/en/model_doc/modernbert
- H100 ModernBERT-base observation: https://www.linkedin.com/posts/michael-feil_the-latest-release-of-infinity-https%3Factivity-7280971190632943616-E07N
- OpenAI Sol pricing: <https://developers.openai.com/api/docs/models/gpt-5.6-sol>
- OpenAI Luna pricing: <https://openai.com/index/advancing-the-price-performance-frontier-with-gpt-5-6/>
- Model cards: <https://huggingface.co/MoritzLaurer>
- GLiClass Modern Large v3: <https://huggingface.co/knowledgator/gliclass-modern-large-v3.0>
- GLiClass Large v3: <https://huggingface.co/knowledgator/gliclass-large-v3.0>